Does Hierarchical Summarization Beat Flat Truncation on Long-Form Fiction?

hotr@itu.dk
IT University of Copenhagen

Abstract

An abstractive summarization models such as BART (Lewis et al., 2020) cap their input at 1024 tokens, forcing a choice when summarizing chapters that exceed this limit: truncate the input, or split it into chunks and summarize hierarchically. We compare flat truncation against a chunk-then-combine hierarchical approach on 40 chapters from the BookSum corpus, using `facebook/bart-large-cnn` without fine-tuning. Across ROUGE-1/2/L and a proper-noun entity-recall check, we find no statistically significant difference between the two approaches using paired Wilcoxon with all $p > 0.44$, with effect sizes shrinking toward zero as the sample grew from 8 to 40 chapters, pretty much signify a genuine null rather than an underpowered test. We compare two mechanisms: reference summaries that emphasise early-chapter content truncation, and by hierarchical summarization's multi-pass compression, visible in its lower entity recall.

1 Introduction

Long-form serialized fiction (web novels, light novels) poses a practical summarization problem: chapters are long, plot-dense, and readers who fall behind want to catch up without any long, boring re-reading sessions. Or any "readers" with a short attention span might preferred summary content instead. Sometimes, model have a certain token limit, hence we explore two differents mechanism to cope with any tokens limit and compare them, hierarchical summarization and flat truncation. We came up with the key question: "Does hierarchical (chunk-then-combine) summarization beat flat truncation on Long-Form Fiction".

2 Method

We use `facebook/bart-large-cnn`, an abstractive encoder–decoder summarizer, loaded directly via `AutoModelForSeq2SeqLM`. (Wolf et al., 2020) BART's encoder accepts at most 1024 tokens; because the transformer self-attention mechanism scales quadratically with sequence length (Vaswani et al., 2017), this limit is architectural rather than incidental, and will be the constraint our comparison is built directly on.

Flat truncation (baseline). The chapter is truncated to the first 1024 tokens and summarized in a single pass; all later content is discarded.

Hierarchical. The chapter is split into ∼1000-token chunks on sentence boundaries, each chunk is summarized independently, the chunk summaries are concatenated, and that concatenation is summarized once more into the final summary.

3 Data

We evaluate on the cleaned BookSum chapter split (Kryściński et al., 2022). BookSum source texts are public-domain novels from Project Gutenberg; reference summaries were collected from web study-guide sources. This is still a proxy for modern web novels. The domain differs, but the structural (narrative input exceeding the model's context window) as well as general long-form fiction format like light novel or web novels is the same. We sample 40 chapters of at least 1500 words so that chapters exceed the 1024-token limit.

4 Evaluation

We report ROUGE-1/2/L F1 (Lin, 2004) against the human reference summary. ROUGE measures lexical n-gram overlap, not factual correctness; we treat it as a proxy and interpret scores relative to the baseline rather than against some certain absolute thresholds. This is because both conditions run on the same chapters, scores are paired; with n small and no way to verify normality of the differences, we use the non-parametric paired Wilcoxon signed-rank test rather than a t-test. As a complement to ROUGE's lexical bias, we also measure entity recall, which is the fraction of proper nouns in the reference that survive into each generated summary.

5 Results

Table 1 reports mean scores at $n = 40$. The two approaches are within ∼0.002 on all three ROUGE

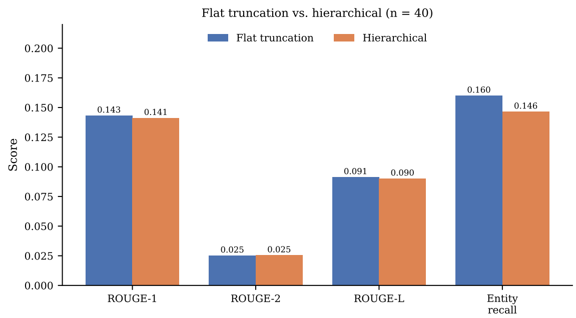

Figure 1: Flat truncation vs. hierarchical across all four metrics ($n = 40$). Scores are near-identical on the three ROUGE metrics; hierarchical is slightly lower on entity recall.

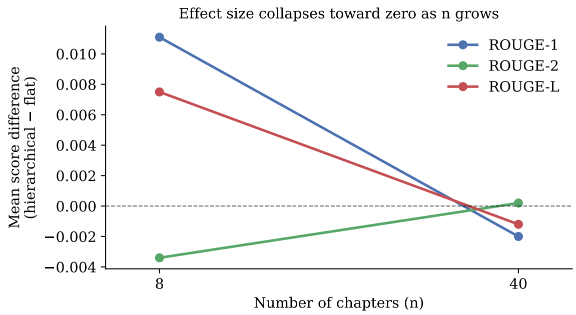

Figure 2: Mean score difference (hierarchical − flat) at $n = 8$ vs. $n = 40$. Differences collapse toward zero as the sample grows, indicating a genuine null rather than an underpowered test.

Metric	Flat truncation	Hierarchical
ROUGE-1	0.1431	0.1411
ROUGE-2	0.0251	0.0253
ROUGE-L	0.0912	0.0900
Entity recall	0.1599	0.1464

Table 1: Mean scores over 40 BookSum chapters. No ROUGE difference is significant (paired Wilcoxon, all $p > 0.44$).

metrics, and the paired Wilcoxon test finds no significant difference on any metric (ROUGE-1 $p = 0.65$, ROUGE-2 $p = 0.46$, ROUGE-L $p = 0.44$). Figure 1 shows the near-identical scores visually.

Crucially, the effect sizes *shrank* as the sample grew from 8 to 40 chapters

(Figure 2): differences that were ± 0.01 at $n = 8$ collapsed toward zero at $n = 40$. A power-limited null would instead show a stable gap that merely failed to reach significance; the shrinkage indicates a *genuine* null.

The one consistent signal is entity recall, where hierarchical is lower in both runs (0.146 vs. 0.160 at $n = 40$), consistent with entities that appear in a single chunk being dropped during the second summarization pass.

6 Discussion and Limitations

Two reasons explain the null. First, hierarchical summarization only helps if the truncation throws away content the reference summary actually cares about. We hypothesise that BookSum reference summaries focus on early-chapter setup — exactly the part truncation keeps — so truncation loses little that ROUGE rewards, leaving hierarchical little to gain. Confirming this would require measuring where reference-relevant content often falls within each chapter. Second, although hierarchical summarization sees more of the chapter, it summarizes twice (even more on higher token data), and each pass loses detail. This shows up in its lower entity recall, where it drops more character and place names. The extra coverage and the lost detail roughly cancel, which is why neither method wins on ROUGE.

Limitations. BookSum is a domain proxy, not actual web novels, so results may not transfer well to modern web-novel style where more ”Internet” term were used. With $n = 40$ the study is modestly powered and detects sizeable effects. ROUGE is purely lexical and cannot credit correct paraphrases, but a semantic metric (e.g. BERTScore) (Zhang et al., 2020) would strengthen the evaluation. We also mention that Entity recall uses a capitalization-based proper-noun finder function rather than a trained NER model, so it over- and under-counts entities, therefore it should only be considered as an estimation.

7 Conclusion

On BookSum chapters, hierarchical summarization does not outperform flat truncation, and is marginally worse at retaining named entities. The benefit of chunking appears to depend on where reference-relevant information sits in the chapter.

References

Wojciech Kryściński, Nazneen Rajani, Divyansh Agarwal, Caiming Xiong, and Dragomir Radev. 2022. BookSum: A collection of datasets for long-form narrative summarization. In *Findings of the Association for Computational Linguistics: EMNLP 2022*, pages 6536–6558, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.

Mike Lewis, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Veselin Stoyanov, and Luke Zettlemoyer. 2020. BART: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension. In *Proceedings of the 58th Annual Meeting of the Association* *for Computational Linguistics*, pages 7871–7880. Association for Computational Linguistics.

Chin-Yew Lin. 2004. ROUGE: A package for automatic evaluation of summaries. In *Text Summarization Branches Out*, pages 74–81, Barcelona, Spain. Association for Computational Linguistics.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. In *Advances in Neural Information Processing Systems 30*, pages 5998–6008.

Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Rémi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, and 3 others. 2020. Transformers: State-of-the-art natural language processing. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 38–45. Association for Computational Linguistics.

Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi. 2020. BERTScore: Evaluating text generation with BERT. In *8th International Conference on Learning Representations (ICLR)*, Addis Ababa, Ethiopia.

8 Contributions and AI Assistance

We used Large Language Models for debugging, code fixing and explanation of complex concepts that was not covered in the curriculum.

We were responsible for every section in this report.